

Checkpoint 2 — Maximum Likelihood Estimator for the Weights

INFO 521 — Machine Learning Foundations

i Course-schedule placement

Sitting: **Week 3** (after Module 3 — Probability, Likelihood & MLE) · retake: Week 4 · gates **Project 1.2**.

Gates: Project 1.2 (Maximum Likelihood) **Format:** in-class, closed-book, ~15–20 min, one derivation. Graded Satisfactory / Not-Yet. **Retakeable.**

Prompt (Version A)

Assume $y_n = \mathbf{w}^\top \phi(x_n) + \epsilon_n$ with $\epsilon_n \sim \mathcal{N}(0, \sigma^2)$ i.i.d. Write the log-likelihood of the dataset and derive the maximum likelihood estimator $\hat{\mathbf{w}}$. State why it coincides with the least-squares solution.

Prompt (Version B — retake)

For the same model, derive the MLE for the noise variance $\hat{\sigma}^2$, treating $\hat{\mathbf{w}}$ as known.

Satisfactory criteria

- Correct dataset likelihood as a product over i.i.d. points; correct log.
- Maximizes (equivalently, drops constants and minimizes the negative log).
- Reaches the stated end result; (A) notes the σ^2 terms are constant in $\mathbf{w}$, so maximizing likelihood = minimizing squared error.

Answer key

A: $\log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w})$. The only $\mathbf{w}$ -dependent term is the squared error, so $\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ — identical to OLS.

B: $\hat{\sigma}^2 = \frac{1}{N} (\mathbf{y} - \mathbf{X}\hat{\mathbf{w}})^\top (\mathbf{y} - \mathbf{X}\hat{\mathbf{w}})$.